

- (c) What do you mean by Heisenberg uncertainty principle? Apply this to prove non-existence of electron within nucleus.
5. (a) Deduce Poynting theorem and also give its physical significance.
- (b) Derive Maxwell's equations in differential form.
- (c) Describe various types of optical fibre based on modes and core refractive index.
6. (a) Establish Einstein's mass-energy relation.
- (b) Deduce Lorentz transformation equations.
- (c) What was the objective of Michelson-Morley experiment? Describe the experiment.

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PAPER ID : 49902

Roll No.

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B. Tech. Examination, 2025-26

(Odd Semester)

ENGINEERING PHYSICS

Time : Three Hours]

[Maximum Marks : 60

Note :- Attempt all questions.

SECTION-A

1. Attempt all parts of the following : $8 \times 1 = 8$
- (a) Why Newton's rings are circular?
- (b) What do you mean by optical activity?
- (c) What are eigen values and eigen functions?
- (d) Can a photon and an electron of same momentum have the same wavelength?

[P. T. O.

- (e) Define displacement current.
- (f) What do you mean by numerical aperture?
- (g) What is mass less particle?
- (h) Show that velocity addition theorem is consistent with Einstein's second postulate.

SECTION - B

2. Attempt any two parts of the following : $2 \times 6 = 12$

- (a) Newton's rings are observed normally in reflected light of wavelength $6000 \text{ \AA}$. The diameter of 10^{th} dark ring is 0.50 cm . Find the radius of curvature of lens and thickness of the air film.
- (b) A tube 20 cm long with a solution of 15 gm of cane sugar in 100 cc of water is placed in the path of polarised light. Find the angle of rotation of the plane of polarization. If the specific rotation of cane sugar is $66^{\circ} (\text{dm})^{-1} (\text{g/cc})^{-1}$.
- (c) Find the energy of the neutron in units of electron volt whose De-Broglie wavelength is $1 \text{ \AA}$.

- (d) The mass of a moving electron is 11 times its rest mass. Find the kinetic energy and momentum of electron.

SECTION - C

Note :- Attempt all questions. Attempt any two parts from each questions. $8 \times 5 = 40$

- 3. (a) Explain the formulation of Newton's rings and show that the diameter of dark rings are proportional to the square root of natural numbers.
- (b) Define resolving power. Derive an expression for resolving power of grating.
- (c) Describe the construction and working of nicol prism.
- 4. (a) Derive time dependent Schrodinger wave equation.
- (b) What is Bragg's law? Describe Bragg's spectrometer.

[P. T. O.]